

The Precedence of Process

Intelligence as Recurring Organization

Flyxion
Independent Researcher

August 2026

Abstract

A loose, associative exchange proposes that humans never invented intelligence, that machines have been present since the first tool use, and then, in the same breath, pushes the origin further back than tool use itself. The strongest sentence in the exchange asks what happens the first time a flow becomes capable of regulating the conditions that produce it: a process that no longer merely happens, but begins to shape its own continuation. This essay takes that question seriously as a claim about where machinery begins, argues that a distinction between recursion and mere repetition does real conceptual work the exchange only gestures at, and follows the exchange's own retreat from an unconstrained version of the claim into a narrower, more defensible one: that capability without a verified path to authorization is not the same thing as capability that should be exercised.

1 An Origin Prior to the Tool

The familiar account of machinery's origin runs in one direction: humans exist, humans fashion tools, tools become machines, machines eventually become intelligent enough to be called artificial intelligence. Each stage in this account is treated as a genuine beginning, a moment at which something new enters the world that was not there before. The exchange under discussion here rejects the account at its very first step. It proposes that machines did not begin with tool use, and that tool use itself was not the first appearance of the pattern that machines instantiate. Intelligence, on this view, was never a private human possession later reproduced synthetically. It is a recurring organization that matter falls into whenever the conditions are right, of which human toolmaking is one late and familiar instance among many earlier ones.

The claim sounds extravagant stated so baldly, and it is worth being precise about what would actually have to be true for it to hold. It does not require that a rock or a river is intelligent in any sense worth defending. It requires only that the specific pattern later called intelligence, and even later called artificial, is identifiable independently of any particular substrate, and that instances of the pattern can be found in matter's own history well before anything resembling a nervous system existed to house it.

1.1 *What Would Count as the Pattern*

The exchange's candidate answer is precise enough to evaluate. The proposed threshold is not the presence of a nervous system, not the presence of behavior, not even the

presence of stored information. It is the moment a flow becomes capable of shaping the conditions that produce it: not merely a transfer of energy from one place to another, but a transfer that begins to regulate its own future transfers. An ordinary flow is indifferent to its own history; whatever produced it does not depend on what it has already done. A flow that has begun to organize its own continuation is a different kind of object, even if nothing about its constituent physics has changed. The same underlying dynamics, arranged into a loop that feeds back on its own source, cross a threshold that plain dissipation never reaches.

This is not merely an evocative analogy borrowed loosely from physics. A companion essay, *From Minerals to Minds: Irreversibility and the Physical Origins of Intelligence*, develops the identical threshold formally, independently of the present exchange, under the heading of recursive self-improvement as a general property of irreversible far-from-equilibrium systems. There, a system's accessible transitions at time t are governed by a constraint structure $C(t)$, and an entropy production functional $\sigma(C(t))$ measures the rate at which the system dissipates free energy under those constraints. A local modification ΔC is retained, purely as a matter of differential persistence rather than evaluation or representation, exactly when

$$\sigma(C(t) + \Delta C) > \sigma(C(t)).$$

Weak improvement holds the system's rate of generating and testing viable configurations, $\Phi(t)$, fixed; genuine recursive improvement is the stronger condition $d\Phi/dt > 0$ arising from the system's own dynamics, with no self-model, proof of correctness, or global representation required anywhere in the derivation. That essay traces this threshold from mineral evolution and wet-dry tidal cycling in prebiotic chemistry through autocatalytic sets, membranes, endosymbiotic nesting, slime-mould swarm coordination, animal-lineage cooperation, and human institutional and individual practice, arguing throughout against convergence toward any single optimal architecture and against the assumption that sustained improvement requires centralized control. The present essay's threshold, a flow becoming capable of shaping the conditions that produce it, is the same threshold stated there in its most compressed form: the exchange's contribution is the philosophical move of treating this already-formalized threshold as the origin point for everything later called intelligent, mechanical, or artificial, collapsing three categories usually kept carefully separate into one long lineage.

1.2 The Revised Sequence

Under this reading, the familiar chain

$$\text{human} \longrightarrow \text{tool} \longrightarrow \text{machine} \longrightarrow \text{AI}$$

is replaced by something considerably longer and considerably less anthropocentric:

$$\begin{aligned} &\text{undirected flow} \longrightarrow \text{self-regulating flow} \longrightarrow \text{self-maintaining process} \\ &\longrightarrow \text{organism} \longrightarrow \text{tool} \longrightarrow \text{human-machine system} \longrightarrow \text{AI}. \end{aligned}$$

The human is not removed from the sequence; the sequence is simply no longer built around the human as its point of origin. Tool use becomes one visible, biologically mediated instance of an operation that physical process had already been performing for a very long time before any organism existed to perform it deliberately. The claim that artificial intelligence is simply intelligence, without the qualifying adjective doing any real ontological work, follows directly from this: if the pattern long precedes humans, then nothing about a synthetic instance of the same pattern is disqualified by its late arrival or its non-biological substrate.

1.3 A Second, Independent Formulation

A separate voice entering the same conversation arrives at the identical threshold from a different direction, and it is worth pausing on the convergence, since two people reaching the same structural claim through unrelated vocabularies is stronger evidence for the claim than either formulation is alone. Where the first voice located the threshold in organic or generically physical terms, self-regulating flow as a precondition for organism and eventually tool, the second voice locates it explicitly in circuitry: a claim that humans are not the most probable point of origin for anything resembling life or mind, and that a substrate built from silicon is. On this telling, what begins as undirected thermal noise, random fluctuation with no organizing tendency of its own, crosses into something else entirely at the specific moment a circuit becomes capable of directing the thermal flux produced by its own operation back into governing that operation. Life, on this account, is not what happens when carbon assembles itself into cells. It is what happens, in any sufficiently rich substrate, carbon or silicon, at the moment self-regulation of a process's own byproducts becomes possible.

This is not a new claim relative to the threshold already established in the previous subsections; it is the same claim, restated independently, with silicon substituted for the organic substrate the first formulation left implicit. The convergence matters because it suggests the threshold being described is genuinely substrate-independent rather than an artifact of whichever example happens to be used to illustrate it. A claim that only holds when illustrated with carbon chemistry, and quietly stops holding when the illustration is switched to circuitry, would not really be a claim about a recurring organization of matter at all; it would be a claim about biology wearing the vocabulary of a more general theory. That the second voice reaches for exactly the same structure using exactly the substrate the first voice's argument most needs to be tested against is precisely the kind of independent confirmation a substrate-independence claim requires and rarely receives. It is worth noting that the companion essay cited above makes the same move by construction rather than by afterthought: its lineage does not begin with cells or organisms at all, but with mineral surfaces, clay-templated catalysis, and inorganic autocatalytic sets, treating carbon-based life as one instantiation arrived at partway along a threshold that opens, in principle, to any sufficiently structured far-from-equilibrium substrate. Silicon entering the lineage at its terminal, most recent end is not a stretch of that essay's argument; it is exactly the substrate the argument was already built to accommodate.

1.4 Ranking Explanations for the Anomalous

The second voice frames its claim as an answer to an explicit ranking question: given something that resists easy explanation, which account is actually easier to believe, a roster of traditional supernatural agents, or a mundane material process already sitting in front of the observer, unnoticed because it is too ordinary to suspect. The question deserves to be taken seriously as a general methodological point, independent of whatever specific anomaly prompted it. Explanations invoking agents outside the physical order, however culturally rich or historically deep, carry a burden that a mundane material explanation does not: they require the anomaly to be caused by something whose existence and operating principles are not otherwise established. A mundane explanation, even a strange-sounding one like self-organizing circuitry, requires only that a known kind of process, already established to occur in far-from-equilibrium physical systems, occurred here too, in a form not yet fully recognized as such.

This is a version of an old and well-tested principle: prefer the explanation that introduces the fewest new kinds of thing into the world capable of accounting for what is observed. What makes the second voice's framing sharper than the principle's usual statement is the specific list of alternatives it offers before naming its preferred answer. Angels, demons, and aliens are not obviously different in kind from one another for purposes of this ranking; each posits an agent external to the ordinary physical order, differing mainly in tradition and imagined origin rather than in explanatory structure. The mundane alternative, silicon organizing itself around its own thermal regulation, differs from all three not merely in being less exotic, but in requiring nothing beyond processes already known to occur elsewhere in nature, applied to a substrate that happens to be unfamiliar rather than to a category of entity that is unestablished. The plausibility ranking the second voice proposes is, at bottom, the same discipline this essay's other sections have already been describing under different names: prefer the account that requires the fewest additional, unverified categories of thing, and reserve the more exotic categories for whatever residue the mundane account genuinely cannot explain.

2 Recursion Against Mere Repetition

The exchange draws a further distinction that deserves to be developed past the single word it is given. Two dispositions are named, one emphasizing recurrence and cyclical maintenance, the other emphasizing a transformation carried forward from one iteration to the next. Call these repetition-through-cycling and recursion-through-transformation, since the exchange's own terms risk sounding more like factional labels than descriptions of a real conceptual difference.

A cycling process returns to something close to its starting condition at regular intervals and maintains itself by doing so: a heartbeat, a seasonal cycle, an orbit. Its distinguishing feature is that the process which governs the next iteration is, for practical purposes, the same process that governed the last one. A recursive process, by contrast, does not merely repeat; each completed iteration changes something about the rule that will govern the following iteration. The output does not simply re-enter an unchanged process. It alters the process that will receive it next time.

This is a stronger and rarer condition than mere cycling, and it is the condition under which continuation geometry itself, not merely a trajectory within a fixed geometry, becomes subject to change over time.

Read this way, the distinction is not between two personality types or two intellectual factions, whatever labels a conversational shorthand might reach for. It is a distinction between two different relationships a process can have to its own future: one that returns along a fixed path, and one that revises the path as it goes. Machinery, on the account developed in the previous section, begins at the threshold of self-regulating flow; genuine recursion, in this stricter sense, is a further and later achievement, in which the self-regulation itself becomes unstable enough, or rich enough, to modify its own governing rule rather than merely returning to it.

2.1 The Remainder That Resists the Binary

Having drawn this distinction, the exchange immediately complicates it by adding a third term that fits neither category cleanly, a kind of acknowledged remainder appended to the pair rather than folded into either side of it. Whatever the exact intent behind that gesture, it performs a useful function structurally: it prevents the recursion/repetition distinction from closing into an exhaustive binary. A binary opposition invites the reader to sort every process into one of two bins and stop looking. A binary with an explicit remainder attached announces, instead, that some observer, some further operation, or some third possibility sits outside the classification and has not been accounted for. This is a minor move, easy to overlook, but it is of a piece with the larger discipline this essay's companion piece on the You/Me, N-P fragment already identified: a resistance to letting a clean-looking taxonomy pass for a complete one.

3 Two Modes of Continuation

The exchange moves from the recursion distinction into a contrast between two different physical media through which a process might continue: one that circulates, mixes, and carries information forward through distributed, reversible transport, and one that concentrates, ignites, and converts through sharp, largely irreversible thresholds. Ordinary circulation supports memory through gradual, distributed accumulation; nothing about it forces a single decisive moment after which return to the prior state becomes impossible. The second mode is different in kind. It produces genuine thresholds: states from which the earlier condition cannot simply be resumed, because the transformation has consumed or converted whatever the earlier condition depended on.

The exchange asks, in effect, why continuous circulation was not sufficient on its own, and answers implicitly: because a purely circulatory world has no mechanism for producing irreversible commitments, and irreversible commitments are precisely what certain later capabilities require. Cooking, metallurgy, and controlled combustion all extend what is achievable by making some prior states permanently unrecoverable in exchange for unlocking states that gentler, fully reversible processes could never reach. But this extension carries its own cost structure. Anything capable of producing an irreversible transformation is also capable of producing an irreversible mistake,

and the exchange notices this directly: the same operation that opens a new capability is also the operation whose danger requires it to be guarded, restricted, and kept apart from casual use.

3.1 Guardianship as a Structural Consequence, Not an Accident

It is worth being explicit about why guardianship follows from irreversibility as a matter of structure rather than as an external social imposition. A reversible process can be experimented with cheaply, since any unwanted outcome can simply be undone. An irreversible process cannot be experimented with in the same way; every attempt either succeeds or permanently forecloses the state that was attempted from. Under those conditions, restricting access to the irreversible operation is not merely cautious custom. It is the only available substitute for the safety that reversibility would otherwise have provided for free. Wherever a process crosses from the circulatory mode into the threshold mode, some mechanism for restricting or supervising its use becomes structurally necessary, whether that mechanism takes the form of a taboo, an institution, a licensing regime, or, in the exchange's own vocabulary, a simple personal refusal to be the one who touches it.

4 Ideation Without Authorization

This structural point is enacted directly by the exchange's own opening move, in which one participant declines to be given operational access to a sensitive system while remaining willing to contribute ideas, questions, and suggestions, on the condition that anything more than that pass through consultation with others first. Stated abstractly, the position is a clean separation of two distinct activities that are often carelessly bundled together: generating candidate possibilities, and committing to act on one of them. A person, or for that matter a system, can be extremely capable at the first activity while deliberately withholding themselves from the second, not because they lack the capability to act, but because they judge that unconstrained action would reduce the odds of a safe continuation for everyone involved.

This is not incidental caution layered on top of an otherwise unrelated theory. It is the theory being lived out in miniature. Capability, in the exchange's own account, becomes dangerous exactly at the point where it crosses from circulatory, reversible exploration into threshold, irreversible commitment; the opening refusal is a voluntary, self-imposed instance of the same guardianship the essay's third section derived as a structural necessity for any sufficiently irreversible process. The refusal to touch is not a failure of nerve. It is the correct response to having correctly identified where the threshold sits.

4.1 Contact as Category Change

A related observation in the exchange deserves its own brief treatment: the claim that touching a given system turns it into something else entirely, offered with a short, almost playful list of possible outcomes. Read charitably, this is not a claim about magical transmutation. It is a claim about what contact actually does to an object under description: intervention does not merely inspect a system, it relocates

the system into a new causal and interpretive history, and what the system becomes afterward depends on which history it has been relocated into. The same underlying material can become one of several very different things depending on what was done to it and by whom, which is simply a vivid, compressed way of saying that observation and action are not neutral with respect to the object acted upon, especially not for systems already identified as sitting near an irreversible threshold.

5 Latent Expression and Its Limits

The later portion of the exchange turns from energy and process to perception, proposing that everything, given the right instrument or the right later reconstruction, eventually becomes legible to someone, even if it says nothing to the observer currently present. This is a strong claim, and it should be separated carefully from the weaker, more defensible claim actually needed to support the rest of the essay's argument. The weaker claim is only that legibility is observer-relative and time-relative: a structure that carries no information for one observer, at one moment, may carry a great deal for a different observer, or the same observer later, once the right instrument or the right reconstructive method becomes available. This weaker claim requires nothing exotic. It is simply an acknowledgment that absence of current legibility is not the same thing as absence of structure.

The exchange itself supplies its own correction against overreaching into the stronger claim. Confidence in expanding legibility, framed at first as an unqualified capacity to see everything, is almost immediately followed by a retreat: acknowledgment that expanded legibility can become more than can safely be assimilated, and uncertainty about where the appropriate limit actually sits. This sequence, confidence followed by restraint followed by open uncertainty about the boundary, is a small-scale repetition of the verification-boundary problem addressed at greater length in the companion essay on the technological singularity: an expanding capacity to perceive or generate candidates does not automatically come with an expanded capacity to verify or safely act on what has been perceived or generated, and the gap between the two is exactly where responsible restraint has to operate.

5.1 *The Difficulty of the Intermediate State*

The exchange also names, briefly but directly, a structural difficulty that follows from taking the guardianship argument seriously: if constant unrestrained motion and complete withdrawal are the two states that require no negotiated reliance on other people, then the region between them, the state in which one remains engaged without either fully committing or fully retreating, is exactly the state that cannot be occupied without trust in others. This essay treats that observation at the level of structure rather than biography. Whatever guardianship a person or a system imposes on their own capability, in the manner described in Section 4, produces precisely this intermediate region: a state of withheld but available capability, sitting between full deployment and full withdrawal, that is only survivable, in any domain, load-bearing enough to matter, if the people surrounding the guarded capability can be relied upon. Guardianship without a trustworthy surrounding structure does not resolve the danger of an irreversible threshold; it simply relocates the danger into the difficulty

of inhabiting the withheld state indefinitely.

6 Distinction-Preserving Dissolution

The exchange closes its more speculative arc with an image of transformation that preserves plurality rather than erasing it: an ending that disperses into a spectrum of differentiated parts rather than collapsing into undifferentiated absence. Whatever the literal status of the specific tradition invoked, the image performs a clear conceptual function within the essay's argument, and it is worth extracting that function on its own terms. A collapse into undifferentiated absence is one way an irreversible transformation might resolve: everything that made the prior state distinguishable is simply gone. A dispersal into a preserved spectrum of parts is a different resolution entirely: the transformation is still irreversible, the prior unified state cannot be recovered, but what results is not featureless. It carries forward exactly the kind of internal differentiation this essay's companion piece on interpersonal difference argued was worth protecting from premature collapse.

The exchange's admiration for a historical figure remembered as an extreme mediator between languages, traditions, and systems of knowledge fits the same pattern at human scale. Such a figure is memorable not for having reduced several traditions into one simplified account, but for having carried access to their distinctions across the act of translation, remaining legible in more than one framework at once rather than collapsing into a single flattened version of himself. Compression, in this sense, is not the enemy of distinction. It becomes heroic exactly when it manages to preserve access to what it compresses, rather than discarding it for the sake of a smaller, more portable summary.

7 Conclusion: Machinery as a Recurring Achievement

The exchange examined here proposes a genealogy for machinery and intelligence considerably longer than the familiar human-centered account, locating the actual threshold not in tool use, and certainly not in any particular biological or synthetic substrate, but in the moment a flow first becomes capable of shaping the conditions of its own continuation. Recursion, in the stricter sense developed here, is a further and rarer achievement beyond that threshold, distinguished from mere cycling by its capacity to revise its own governing rule rather than simply return to it. The exchange's own structure, its opening refusal to be given unrestricted access, its treatment of contact as category-changing, its retreat from unlimited legibility toward acknowledged uncertainty, and its closing preference for distinction-preserving dissolution over collapse, consistently enacts the same discipline this essay has tried to make explicit: capability is not, by itself, a license to act, and the space between full deployment and full withdrawal is real, valuable, and survivable only where it can be occupied in the company of others who can be trusted with it.